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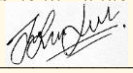
Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – TECHNICAL PRESENTATION

Course Code:	ITA03	**Course Title:**	Mobile computing
CO Assessed:	CO3 – Precision (BL3)	**Assessment:**	Assessment Tool 2 – Technical Presentation
Weightage:	30 % of CIA	**Total Marks:**	20 (Answer ANY ONE)
CO3 Statement:	*Implement wireless networking and Mobile IP concepts for routing in cellular and ad hoc networks.. (BTL 3)*		
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Section / Batch:	Slot A	**Date:**	23-10-26

Code of Conduct

I Joel Benn John B (1911260003) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: 

Instructions

- Duration: 10 minutes

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Questions

1. Derive the processing gain for a system with a 3.6864 Mcps chip rate at three different data rates — 9.6, 19.2, and 38.4 kbps — showing all three calculations clearly side by side. Comment on the trend you observe (processing gain halves as data rate doubles), and discuss the practical implication this has for a network trying to support both voice and higher-rate data services on the same infrastructure.

Processing Gain in a 3.6864 Mcps CDMA System

Derivation, trend analysis and implications for mixed voice and data networks

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1. Introduction and Problem Statement

Code Division Multiple Access (CDMA) lets many users transmit at the same time in the same frequency band. Each user's data is multiplied by a high-rate spreading code, so the transmitted signal occupies a bandwidth far larger than the data itself. At the receiver the same code is applied again, which collapses the wanted signal back to its original narrow bandwidth while the interference stays spread out. The benefit obtained by this spreading and despreading is called the **processing gain** (G_p), and it is one of the most important design parameters of a CDMA network.

Problem. A CDMA system uses a fixed chip rate of R_c = 3.6864 Mcps (million chips per second). Derive the processing gain at three data rates, 9.6 kbps, 19.2 kbps and 38.4 kbps, and present the three calculations side by side. Then comment on the trend (the processing gain halves each time the data rate doubles) and discuss what this means for a network that must carry both voice and higher-rate data on the same infrastructure.

Objectives of this answer

- Define processing gain and derive it from first principles, in both linear form and decibels.
- Compute and verify the value for each of the three data rates, with the supporting bit and chip durations.

- Explain the inverse relationship between data rate and processing gain, including the 3.01 dB per doubling rule.
- Show quantitatively how a lower processing gain affects required power, cell capacity and coverage, and describe how networks cope.

2. Background: Spread Spectrum and CDMA

2.1 Direct-sequence spreading

In a direct-sequence CDMA (DS-CDMA) transmitter each information bit, which lasts T_b seconds, is multiplied by a pseudo-random noise (PN) code made up of N much shorter pulses called **chips**, each lasting T_c seconds. Because there are N chips in every bit, the chip rate is N times the bit rate. Table 1 lists the symbols used throughout this answer.

Table 1: Key terms and symbols

Term	Symbol	Meaning
Data (bit) rate	R_b	Number of information bits per second, here 9.6, 19.2 or 38.4 kbps
Chip rate	R_c	Number of code chips per second, fixed at 3.6864 Mcps
Bit duration	$T_b = 1/R_b$	Time taken by one information bit
Chip duration	$T_c = 1/R_c$	Time taken by one chip, about 271.27 ns here
Spreading factor	N	Chips per bit, $N = T_b/T_c = R_c/R_b$
Processing gain	G_p	Ratio of spread bandwidth to information bandwidth, equal to N for ideal spreading
Energy per bit	E_b	Received signal energy in one bit, $E_b = S \cdot T_b$
Interference density	I_0 or N_0	Interference plus noise power per hertz

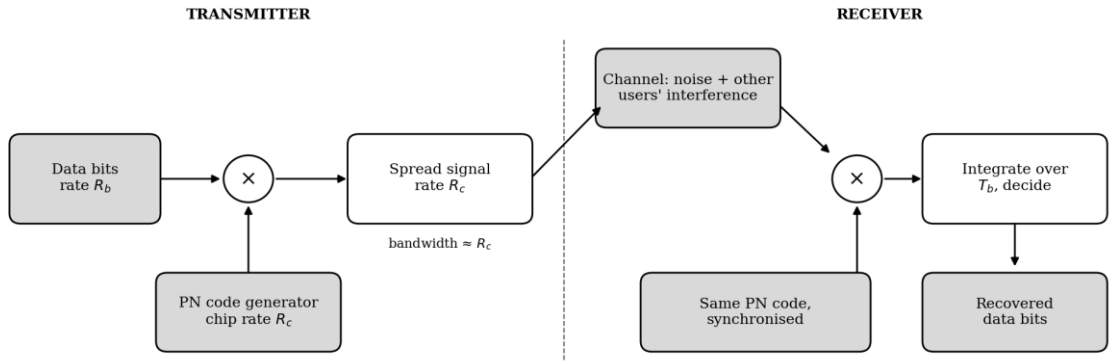


Figure 1: Block diagram of direct-sequence spreading and despreading. The wanted signal is correlated with the same synchronised PN code at the receiver.

2.2 Why spreading helps

Figure 2 shows the idea in the frequency domain. Spreading takes the same signal power and spreads it over a bandwidth roughly R_c, so the power spectral density drops by a factor of about G_p. In the channel the signal is mixed with noise and with other users' signals. At the receiver, multiplying by the synchronised code re-concentrates the wanted signal into a narrow bandwidth of about R_b. Signals that are not correlated with the code, such as other users' transmissions and noise, are not compressed. They remain wide and low, so a narrow filter or integrator after despreading passes only about $1/G_p$ of their power.

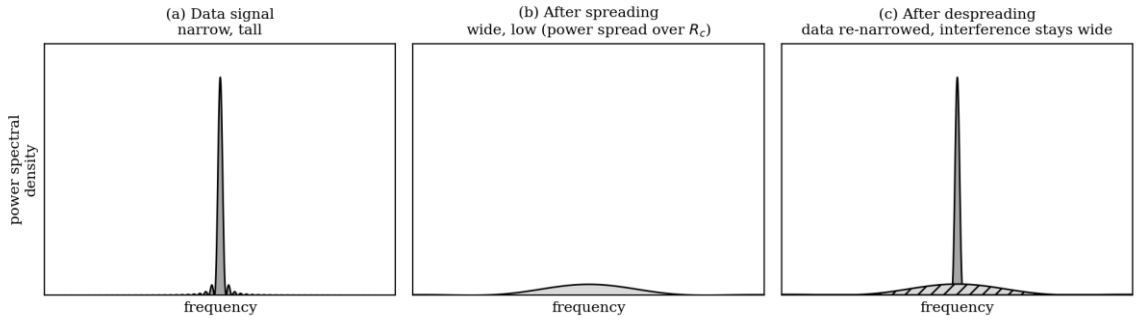


Figure 2: Spectra at each stage (schematic, with an exaggerated gain of about 20 for visibility). The shaded hatched region is interference.

2.3 The 3.6864 Mcps chip rate

A chip rate of 3.6864 Mcps equals exactly three times 1.2288 Mcps, the chip rate of IS-95 and of cdma2000 1x. It is the chip rate of the wider-band (3x) option of cdma2000. For reference, 1.2288 Mcps with a 9.6 kbps voice channel gives a spreading factor of 128, or 21.07 dB. At 3.6864 Mcps the same 9.6 kbps channel gets 384, which is three times larger and $10 \log_{10} 3 \approx 4.77$ dB better. This is the main reason to widen the bandwidth when higher data rates are wanted (see Section 6.6).

3. Definition and Derivation of Processing Gain

3.1 From bandwidth and from timing

Processing gain is defined as the ratio of the transmitted (spread) bandwidth to the information bandwidth:

$$G_p = \frac{W_{\text{spread}}}{W_{\text{info}}}$$

The bandwidth of a spread-spectrum signal is proportional to its chip rate, and the bandwidth of the information signal is proportional to its bit rate, with the same constant of proportionality. The two bandwidths therefore scale like the two rates. The same result follows from timing:

1. One bit lasts $T_b = 1/R_b$ seconds and one chip lasts $T_c = 1/R_c$ seconds.
2. The number of chips in one bit is $N = T_b/T_c$.
3. Substituting gives $N = (1/R_b)/(1/R_c) = R_c/R_b$.

4. For ideal spreading the processing gain equals the spreading factor, so $G_p = N = R_c/R_b$.

$$G_p = \frac{R_c}{R_b} = \frac{T_b}{T_c}$$

In decibels, because the logarithm turns a ratio into a difference:

$$G_p\,(\mathrm{dB}) = 10\log_{10}\left(\frac{R_c}{R_b}\right) = 10\log_{10} R_c \;-\; 10\log_{10} R_b$$

3.2 Link to E_b/N_0 and the required signal-to-interference ratio

Processing gain matters because it connects the signal-to-interference ratio at the antenna to the bit-level quality that decides the error rate. Let the received signal power be S and the total wideband interference plus noise power be I, spread roughly evenly over a bandwidth R_c.

1. Energy per bit is the signal power times the bit duration: $E_b = S \cdot T_b = S/R_b$.
2. Interference power spectral density is $I_0 = I/R_c$.
3. Dividing gives $E_b/I_0 = (S/R_b) \,/\, (I/R_c) = (S/I)\cdot(R_c/R_b) = (S/I)\cdot G_p$.

$$\frac{E_b}{I_0} = \frac{S}{I}\cdot G_p \quad \Rightarrow \quad \left(\frac{S}{I}\right)_{\mathrm{req}} = \frac{(E_b/I_0)_{\mathrm{req}}}{G_p}$$

A given service needs a roughly fixed E_b/N_0 to reach its target error rate. The required pre-despreading S/I is therefore inversely proportional to G_p. In decibels, $(S/I) = (E_b/I_0) - G_p$, which is how a negative S/I (signal below the interference) can still give a usable link.

3.3 Interference margin

The amount of interference a link can tolerate is called the interference (or jamming) margin:

$$M_j\,(\mathrm{dB}) = G_p\,(\mathrm{dB}) \;-\; (E_b/N_0)_{\mathrm{req}}\,(\mathrm{dB}) \;-\; L_{\mathrm{sys}}\,(\mathrm{dB})$$

Here L_{sys} covers implementation losses. A higher processing gain gives a larger margin. This is the quantity that shrinks when the data rate goes up.

4. Calculation at 9.6, 19.2 and 38.4 kbps

4.1 Given data

- Chip rate R_c = 3.6864 Mcps = 3.6864×10^6 chips/s (fixed for all three cases).
- Data rates R_b = 9.6 kbps, 19.2 kbps and 38.4 kbps, that is 9.6×10^3, 19.2×10^3 and 38.4×10^3 bits/s.
- Formulae: $G_p = R_c/R_b$ and $G_p(\mathrm{dB}) = 10\log_{10}(G_p)$.

4.2 Worked calculations

Case 1: R_b = 9.6 kbps

1. $G_p = (3.6864 \times 10^6) / (9.6 \times 10^3) =$ **384**.
2. G_p(dB) = 10 $\log_{10}(384)$. Since $\log_{10}(384) = \log_{10}(3.84) + 2 = 0.5843 + 2 = 2.5843$, $G_p =$ **25.84 dB**.
3. Check: $T_b = 1/9600 = 104.17$ µs and $T_c = 1/(3.6864 \times 10^6) = 271.27$ ns, so $T_b/T_c = 384$ chips per bit.

Case 2: R_b = 19.2 kbps

1. $G_p = (3.6864 \times 10^6) / (19.2 \times 10^3) =$ **192**.
2. G_p(dB) = 10 $\log_{10}(192) = 10 \log_{10}(384) - 10 \log_{10}(2) = 25.84 - 3.01 =$ **22.83 dB**.
3. Check: $T_b = 1/19200 = 52.08$ µs, giving 52.08 µs / 271.27 ns = 192 chips per bit.

Case 3: R_b = 38.4 kbps

1. $G_p = (3.6864 \times 10^6) / (38.4 \times 10^3) =$ **96**.
2. G_p(dB) = 10 $\log_{10}(96) = 10 \log_{10}(384) - 10 \log_{10}(4) = 25.84 - 6.02 =$ **19.82 dB**.
3. Check: $T_b = 1/38400 = 26.04$ µs, giving 26.04 µs / 271.27 ns = 96 chips per bit.

4.3 Side-by-side comparison

Table 2: Processing gain at a fixed chip rate of 3.6864 Mcps

Quantity	Case 1	Case 2	Case 3
Data rate R_b	9.6 kbps	19.2 kbps	38.4 kbps
Chip rate R_c	3.6864 Mcps	3.6864 Mcps	3.6864 Mcps
$G_p = R_c/R_b$	$3.6864\times10^6 / 9.6\times10^3 =$ **384**	$3.6864\times10^6 / 19.2\times10^3 =$ **192**	$3.6864\times10^6 / 38.4\times10^3 =$ **96**
G_p(dB)	10 $\log_{10}(384) =$ **25.84 dB**	10 $\log_{10}(192) =$ **22.83 dB**	10 $\log_{10}(96) =$ **19.82 dB**
Chips per bit N	384	192	96
Bit duration T_b	104.17 µs	52.08 µs	26.04 µs
Chip duration T_c	271.27 ns	271.27 ns	271.27 ns
Change from Case 1	reference	÷ 2 (−3.01 dB)	÷ 4 (−6.02 dB)

4.4 Verification

Because $G_p = R_c/R_b$, the product $G_p \times R_b$ must always equal the chip rate. The three results satisfy this: 384 × 9.6 kbps = 192 × 19.2 kbps = 96 × 38.4 kbps = 3,686,400 chips/s. The data rates are in the ratio 1 : 2 : 4 and the gains in the ratio 4 : 2 : 1, as expected for an inverse relationship.

4.5 Extending the pattern

Continuing to double the data rate gives the values in Table 3. The processing gain falls to only 12 (10.79 dB) at 307.2 kbps, which shows why very high rates cannot be offered on a narrowband carrier without a large capacity penalty.

Table 3: Processing gain at higher data rates (3.6864 Mcps)

Data rate	Gp (linear)	Gp (dB)	Loss vs 9.6 kbps
76.8 kbps	48	16.81	9.03 dB
153.6 kbps	24	13.80	12.04 dB
307.2 kbps	12	10.79	15.05 dB

5. Analysis of the Trend

5.1 Inverse proportionality

With the chip rate fixed, $G_p = R_c/R_b$ is inversely proportional to the data rate. Doubling R_b halves G_p, as the three results show (384 → 192 → 96). Plotted against rate the linear gain traces a hyperbola; plotted in decibels against the logarithm of the rate it is a straight line, because the logarithm turns the repeated halving into equal steps (Figure 3).

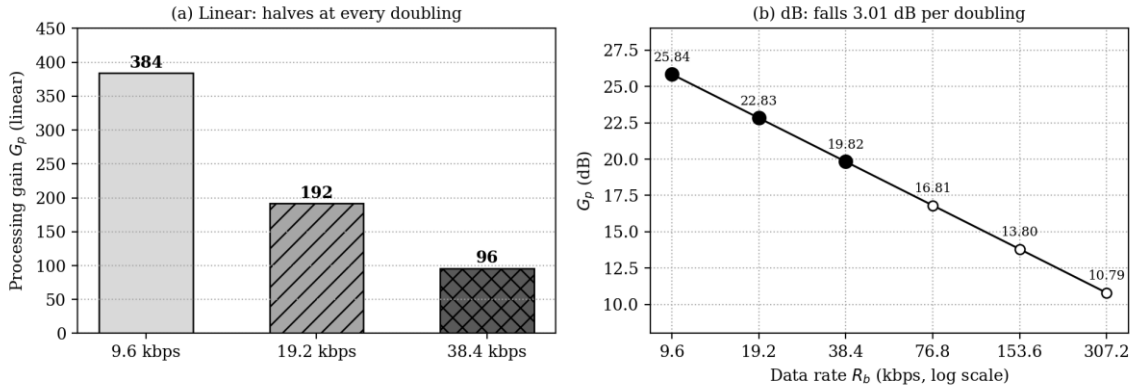


Figure 3: Processing gain versus data rate at 3.6864 Mcps. Filled markers in (b) are the three required cases.

5.2 General relation and the 3.01 dB rule

If the data rate changes from R_{b1} to R_{b2}, the processing gain changes as follows:

$$G_{p2} = G_{p1} \cdot \frac{R_{b1}}{R_{b2}} \qquad \Delta G_p\,(\text{dB}) = -10\log_{10}\left(\frac{R_{b2}}{R_{b1}}\right)$$

For a doubling, $10\log_{10}(2) = 3.01$ dB, so the loss is about 3 dB each time. Table 4 collects the common cases.

Table 4: Effect of rate increase on processing gain

Rate increase	Gain divided by	Change in Gp (dB)
× 2	2	−3.01
× 4	4	−6.02
× 8	8	−9.03
× 10	10	−10.00
× 16	16	−12.04

5.3 Physical interpretation

The loss of processing gain has a simple physical meaning, which can be seen in two equivalent ways.

- **Fewer chips per bit.** At 38.4 kbps each bit is spread over only 96 chips instead of 384. The receiver averages over fewer chips, so it suppresses interference less effectively.
- **Less energy per bit.** Since $E_b = S/R_b$, doubling the rate halves the energy in each bit for the same received power S. To keep E_b/I_0 constant the transmitter must double its power, which is exactly 3 dB.

The chip rate sets a fixed total "budget" of spreading. Giving a user more bits per second spends that budget faster, and the price is paid in power and therefore in interference to everyone else.

5.4 Limits of the idealised model

The figures above are idealised. Real systems use channel coding (which lowers the E_b/N_0 needed), rake receivers, imperfect orthogonality and power-control errors, and spreading factors are normally powers of two in orthogonal-code schemes rather than values like 384. None of this changes the central trend. The required power still scales roughly in proportion to the data rate, because the underlying relationship $E_b = S/R_b$ is unchanged.

6. Practical Implications for Voice and Higher-Rate Data

6.1 Voice and data place different demands on the network

Table 5: Typical voice and higher-rate data services on the same carrier

Property	Voice (9.6 kbps)	Higher-rate data (38.4 kbps)
Processing gain at 3.6864 Mcps	384 (25.84 dB)	96 (19.82 dB)
Relative transmit power for equal E_b/N_0	1×	4× (+6.02 dB)
Delay sensitivity	High, must be real time	Usually lower, can be buffered
Error tolerance	Tolerates occasional frame loss	Needs very low error rate, uses retransmission
Activity pattern	Active about 40 to 50% of the time	Bursty, but bursts carry many bits

6.2 Higher power and a smaller interference margin

Using the relation from Section 3.2 and an illustrative target of E_b/N_0 = 7 dB, the pre-despreading signal-to-interference ratio each link needs is $(S/I)_{req}$ = 7 dB − G_p(dB):

Table 6: Required signal-to-interference ratio and interference margin (E_b/N_0 = 7 dB, no implementation loss)

Data rate	Gp (dB)	Required S/I	Interference margin	Relative power
9.6 kbps	25.84	−18.84 dB	18.84 dB	1× (0 dB)
19.2 kbps	22.83	−15.83 dB	15.83 dB	2× (+3.01 dB)
38.4 kbps	19.82	−12.82 dB	12.82 dB	4× (+6.02 dB)

The 7 dB value is only an assumption for demonstration; the real figure depends on the coding, the channel and the target frame error rate. The pattern is what matters. Each doubling of rate raises the required S/I by 3 dB, cuts the interference margin by 3 dB and doubles the power the user must radiate.

6.3 Capacity: a data user is expensive

In a CDMA cell all users share one band and each user's signal is interference to the others, so capacity is interference-limited. If N users of equal power S are active, a given receiver sees the wanted power S against interference of about $(N-1)S$, giving $S/I \approx 1/N$. Combining with $E_b/I_0 = (S/I)\cdot G_p$ from Section 3.2 gives the ideal single-cell pole capacity:

$$N_{\max} \approx \frac{G_p}{(E_b/N_0)_{\text{req}}}$$

In a multi-cell system, interference from neighbouring cells (a fraction f of own-cell interference, typically around 0.5) lowers this, while voice activity v (about 0.4 to 0.5 for speech) raises it for voice users:

$$N_{\max} \approx \frac{G_p}{(E_b/N_0)_{\text{req}} \cdot \nu \cdot (1+f)}$$

Table 7: Illustrative pole capacity per cell (E_b/N_0 = 7 dB = 5.01)

Data rate	Gp	Ideal, single cell	With other-cell factor f = 0.55
9.6 kbps	384	76.6 users	49.4 users
19.2 kbps	192	38.3 users	24.7 users
38.4 kbps	96	19.1 users	12.4 users

The capacity falls in the same 4 : 2 : 1 ratio. With voice activity v = 0.5 the 9.6 kbps figure rises from 49.4 to about 98.9 voice users, which is the "soft" capacity gain that continuously active data users do not enjoy. A 38.4 kbps user therefore consumes roughly the interference budget of **four** voice users.

Mixed-load example. Networks are normally loaded to only a part of pole capacity (commonly 50 to 75%) to keep noise rise and power-control stability under control. Take the 49.4-user pole capacity and a 50% load, giving a budget of about 24 voice-equivalents, and let each 38.4 kbps data user count as four voice-equivalents. Table 8 and Figure 4(a) show how adding data users displaces voice users.

Table 8: Sharing a budget of 24 voice-equivalents

38.4 kbps data users	Voice-equivalents used by data	Voice users that still fit
0	0	24
1	4	20
2	8	16
3	12	12
4	16	8
5	20	4
6	24	0

6.4 Coverage shrinks for high-rate users

Because a higher-rate link needs more power for the same quality, it reaches the transmitter's maximum power at a shorter distance. This matters most on the uplink, where a handset has a limited power budget. If received power falls with distance as d^{-n}, and the required transmit power rises by ΔP dB, the maximum range changes by the factor

$$\frac{d_2}{d_1} = 10^{-\Delta P/(10n)}$$

With a path-loss exponent of n = 3.5 (assumed for an urban area), a 3.01 dB increase shrinks the radius to 0.82 of its original value, and a 6.02 dB increase shrinks it to 0.67 (Figure 4(b)). The area covered at 38.4 kbps is then only about $0.67^2 \approx 45\%$ of the area covered at 9.6 kbps. The result is **cell breathing**. As load grows, interference rises and the usable cell radius shrinks. If many high-rate users are present, voice users at the cell edge can be dropped even though their own rate is low.

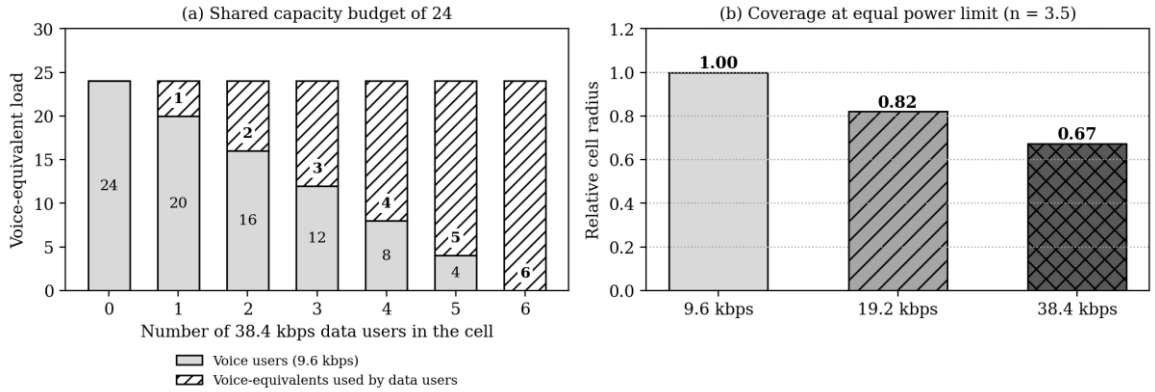


Figure 4: (a) How data users displace voice users within a fixed capacity budget; (b) relative coverage radius when the same power limit applies (n = 3.5).

6.5 Near-far problem and power control

A user that is close to the base station and transmits at the power needed by a distant user would swamp the others. High-rate users, which need the most power, make this worse. CDMA networks therefore rely on **fast closed-loop power control**, updated hundreds of times per second, so that every user arrives at the base station with just enough power for its own E_b/N_0 target and no more. Good power control is the reason the capacity formulas above can be approached in practice.

6.6 How networks support both services together

Since data rate is paid for with processing gain, and so with capacity, power and coverage, a network cannot treat all users alike. The usual measures are:

1. **Variable spreading factor and multicode transmission.** A high-rate user is given a shorter spreading code (lower G_p) or several lower-rate code channels in parallel. Note that neither option removes the cost. A shorter code lowers the gain per bit, while parallel codes add up their powers, so interference is still roughly proportional to the total rate.
2. **Rate adaptation by location and channel quality.** Offer the peak rate near the cell centre where path loss is small, and fall back to lower rates (higher G_p) at the cell edge.
3. **Admission and congestion control.** Reserve capacity for voice, which is delay-critical, and give data the remaining load. Under congestion, reduce data rates first or delay data bursts.

4. **Scheduling and time-sharing of data.** Because data is bursty and delay-tolerant, short high-rate bursts can be scheduled when the cell is lightly loaded, so average interference stays low.
5. **Exploiting voice activity.** Silent periods in speech lower the average interference by about half, creating room for data.
6. **Tight power control.** Prevent any user from radiating more power than necessary, which protects both voice and data users.
7. **Wider bandwidth.** A higher chip rate restores processing gain. Moving from 1.2288 to 3.6864 Mcps triples G_p for any given rate (+4.77 dB), so a 38.4 kbps user at 3.6864 Mcps ($G_p = 96$) is as well protected as a 12.8 kbps user would be at 1.2288 Mcps.
8. **Separate carriers or hierarchical cells.** High-rate data can be placed on its own carrier, or in small cells, so that it cannot erode voice capacity or coverage on the main macro carrier.
9. **Better receivers and coding.** Rake receivers, forward error correction, interleaving and multi-user detection lower the E_b/N_0 needed or remove interference, which gives some of the lost margin back.

6.7 Summary of trade-offs

Table 9: Consequences of a falling processing gain and the usual response

Issue	Effect of higher data rate	Network response
Required power	+3 dB per doubling (2× power)	Fast power control, rate adaptation
Interference margin	−3 dB per doubling	Admission control, load limits
Cell capacity	Data user ≈ 4 voice users at 38.4 kbps	Reserve capacity, schedule bursts
Coverage	Radius ≈ 0.82 then 0.67 of the 9.6 kbps value	Offer top rates only near the cell centre
Voice quality	Can degrade as data load rises	Prioritise voice, use separate carriers
Spreading budget	Fixed by chip rate	Widen bandwidth (higher chip rate)

7. Likely Exam Questions with Answers

Q1. Define processing gain and give its formula.
Processing gain is the ratio of the spread bandwidth to the information bandwidth, which equals the ratio of the chip rate to the bit rate: $G_p = R_c/R_b$, or $10 \log_{10}(R_c/R_b)$ dB.

Q2. Find the processing gain for a 3.6864 Mcps chip rate and 76.8 kbps data.
$G_p = 3.6864 \times 10^6 / 76.8 \times 10^3 = 48$, which is $10 \log_{10}(48) = 16.81$ dB.

Q3. Why does the processing gain fall by 3 dB when the data rate doubles?
The chip rate is fixed, so G_p is inversely proportional to R_b. Doubling the rate halves the gain, and $10 \log_{10}(2) = 3.01$ dB. Physically, each bit has half the energy, so twice the power is needed to keep E_b/N_0 constant.

Q4. How many 9.6 kbps voice users does one 38.4 kbps user equal, and why?
Four. At equal E_b/N_0, the interference contributed by a user is proportional to its power, and the required power is proportional to the rate (38.4/9.6 = 4).

Q5. Name three ways a network can support voice and high-rate data on one carrier.
Variable spreading factor or multicode transmission, admission control that protects voice capacity, and scheduling of bursty data, together with fast power control and a wider chip rate where available.

Q6. What is cell breathing?
Cell breathing is the shrinking of the usable cell radius as load and interference rise. High-rate users make it worse because they need more power and raise interference for all other users.

8. Conclusion, Formula Summary and Suggested Reading

8.1 Conclusion

At a fixed chip rate of 3.6864 Mcps the processing gain is 384 (25.84 dB) at 9.6 kbps, 192 (22.83 dB) at 19.2 kbps and 96 (19.82 dB) at 38.4 kbps. Each doubling of the data rate halves the processing gain and costs 3.01 dB. This is because the processing gain is simply the number of chips used to carry each bit, and the chip rate is a fixed resource.

For a network, this means that data rate is purchased directly with interference margin. A higher-rate user needs proportionally more power, uses up proportionally more of the shared capacity (a 38.4 kbps user counts for about four voice users), and has a smaller coverage radius. Supporting voice and higher-rate data on the same infrastructure therefore needs careful load management, power control, scheduling and, where possible, more bandwidth, rather than treating all users alike.

8.2 Formula summary

Table 10: Key formulae

Quantity	Expression
Processing gain	$G_p = R_c/R_b = T_b/T_c = N$
Processing gain in dB	$G_p(\text{dB}) = 10 \log_{10}(R_c/R_b)$
Effect of rate change	$\Delta G_p(\text{dB}) = -10 \log_{10}(R_{b2}/R_{b1})$
Bit-level SIR relation	$E_b/I_0 = (S/I)\cdot G_p$
Interference margin	$M_j = G_p(\text{dB}) - (E_b/N_0)_{req}(\text{dB}) - L_{sys}(\text{dB})$
Pole capacity	$N_{max} \approx G_p / [(E_b/N_0)\cdot v\cdot(1+f)]$
Range scaling	$d_2/d_1 = 10^{-\Delta P/(10n)}$

8.3 Suggested reading

- T. S. Rappaport, *Wireless Communications: Principles and Practice*, Prentice Hall, chapter on spread spectrum and CDMA.
- A. J. Viterbi, *CDMA: Principles of Spread Spectrum Communication*, Addison-Wesley, 1995.
- K. S. Gilhousen et al., "On the capacity of a cellular CDMA system," *IEEE Transactions on Vehicular Technology*, vol. 40, no. 2, 1991.
- H. Holma and A. Toskala, *WCDMA for UMTS*, Wiley.